

Radical Equations Practice (Algebra II)

Instructions

Solve each equation over the real numbers. Show your work on separate paper and give all solutions in exact form. Check each proposed solution in the original equation; if none work, write “no real solution.” Work without a calculator. Aim to finish in 40 minutes.

Problems

1. $\sqrt{3x+4} = 7$
2. $2\sqrt{x+5} - 3 = 9$
3. $\sqrt[3]{4x-7} + 2 = -1$
4. $\sqrt{5x+6} = x$
5. $\sqrt{2x+9} = x - 3$
6. $\sqrt{4x+5} = x - 1$
7. $\sqrt{3x+7} = \sqrt{x+1} + 2$
8. $\sqrt{x+5} + \sqrt{x-3} = 4$
9. $\sqrt{2x+3} - \sqrt{x-1} = 2$
10. $\sqrt{5-x} = x - 6$
11. $\sqrt{x^2 - 5x + 10} = 2$
12. $\sqrt{x + \sqrt{4x - 4}} = 3$